

CS329A: Agentic AI and Frontier Research

Stanford Frontier AI Learning System

Baseline: October 7, 2026. Print companion to the HTML course site:
course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

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Course map

Unit graph (full course: U01-U08)

U01 Test-time compute and verification

needs P07 (estimation), P14 (transformer), P22 (experiments)
feeds U03 (search uses sampling), U05 (selection for code),
U08 (evaluation of compute tradeoffs)

U02 Feedback, tools, and constitutional learning

needs P17 (RL), P20 (tools), P21 (security)
feeds U03 (tool traces feed train-time RL), U05 (code agents)

U03 Planning, search, and train-time RL

needs P09 (optimization), P17 (RL), P20 (tools)
feeds U04 (search drives evolution), U07 (proof search),
U08 (long-horizon eval)

U04 Open-ended evolution and deep research

needs P20 (tools), P22 (experiments)
feeds U08 (project frame), capstones

U05 Software-engineering and kernel agents

needs P12, P15, P20, P22
feeds U08 (agent eval), capstones
session 13 (Nov 3), Agentic Frameworks for Software Engineering

U06 Memory, caches, and long-context representations

needs P14, P19, P20
feeds U07 (memory for long proofs), U08 (long-horizon tasks)
session 14 (Nov 7), guest Junchen Jiang (LMCache, schedule line only)

U07 Reasoning, formal systems, and autonomy

needs P17, P20, P22
feeds U08 (verifiers and judges), capstones
sessions 15 (Denny Zhou), 16 (Thang Luong), 18 (Misha Laskin),
19 (Danny Driess). Guests at schedule-line level only

U08 Long-horizon evaluation and research projects

needs P07, P22, P24

feeds capstones, open questions

session 17 (Nov 17), Agentic Evaluations and Long-Horizon Tasks

Session mapping to the official Autumn 2025 schedule

Unit	Sessions	Papers anchor
U01	2 (Sep 26), 3 (Sep 29)	repeated sampling, best-of-N, self-consistency, inference architecture search, verification
U02	4 (Oct 3)	ReAct, execution feedback, constitutional AI
U03	5 (Oct 6), 6 (Oct 10)	tree search, decomposition, parallel planning, STaR, reasoning RL, DAPO/GRPO
U04	7 (Oct 13), 8 (Oct 17), 9 guest (Oct 20)	agent architecture search, AI Scientist, AlphaEvolve, AlphaCode, deep research

Unit	Sessions	Papers anchor
U05	13 (Nov 3)	CodeMonkeys serial/parallel scaling, Kernel-Bench, fast_p
U06	14 (Nov 7), guest Junchen Jiang	MemGPT, CacheBlend, Cartridges, LMCache (guest line only)
U07	15 (Nov 10), 16 (Nov 14), 18 (Nov 21), 19 (Dec 1), guests Denny Zhou, Thang Luong, Misha Laskin, Danny Driess	LLM reasoning, AlphaProof/AlphaGeometry, autonomy, robotics (guest lines only)
U08	17 (Nov 17)	GDPVal, DeepScholar-Bench, duration curves, judge contamination, project gates

File map per unit

Unit	Lesson	Keys	Lab	Lab key	Visuals	Interview
U01	lessons/u01_ test_time_ compute.md	keys/ u01_ answers. md	labs/ u01_ lab.md	labs/ keys/ u01_key. md	visuals/render_u01. py, u01_fig01..03. png	interview/u01_ questions.md, u01_key.md
U02	lessons/u02_ feedback_tools. md	keys/ u02_ answers. md	labs/ u02_ lab.md	labs/ keys/ u02_key. md	visuals/render_u02. py, u02_fig01..03. png	interview/u02_ questions.md, u02_key.md
U03	lessons/u03_ planning_ search.md	keys/ u03_ answers. md	labs/ u03_ lab.md	labs/ keys/ u03_key. md	visuals/render_u03. py, u03_fig01..03. png	interview/u03_ questions.md, u03_key.md
U04	lessons/u04_ open_evolution. md	keys/ u04_ answers. md	labs/ u04_ lab.md	labs/ keys/ u04_key. md	visuals/render_u04. py, u04_fig01..03. png	interview/u04_ questions.md, u04_key.md

Unit	Lesson	Keys	Lab	Lab key	Visuals	Interview
U05	lessons/u05_ swe_kernel_ agents.md	keys/ u05_ answers. md	labs/ u05_ lab.md	labs/ keys/ u05_key. md	visuals/render_u05. py, u05_fig01..03. png	interview/u05_ questions.md, u05_key.md
U06	lessons/u06_ memory_caches. md	keys/ u06_ answers. md	labs/ u06_ lab.md	labs/ keys/ u06_key. md	visuals/render_u06. py, u06_fig01..03. png	interview/u06_ questions.md, u06_key.md
U07	lessons/u07_ reasoning_ formal.md	keys/ u07_ answers. md	labs/ u07_ lab.md	labs/ keys/ u07_key. md	visuals/render_u07. py, u07_fig01..03. png	interview/u07_ questions.md, u07_key.md
U08	lessons/u08_ long_horizon_ eval.md	keys/ u08_ answers. md	labs/ u08_ lab.md	labs/ keys/ u08_key. md	visuals/render_u08. py, u08_fig01..03. png	interview/u08_ questions.md, u08_key.md

Capstones: capstones/research_capstone.md, capstones/applied_capstone.md, with pilot scripts superseded to capstones/superseded/pilot_research.py and capstones/superseded/pilot_applied.py.

Prerequisites

Shared bridges live at `../shared/prerequisites/`. Link, do not rebuild. Each unit below names its bridge files plus the local remediation taught inside this course.

U01 Test-time compute and verification

- `../shared/prerequisites/p07_estimation.md`: sampling, estimators, standard

errors, confidence intervals, calibration. Local remediation: lesson C01 re-derives `pass@N` from Bernoulli sampling before any curve is drawn.

- `../shared/prerequisites/p14_transformer.md`: next-token sampling,

temperature, top-p. Local remediation: lesson C01 defines a sample as one full decoded completion with a fixed decoding policy.

- `../shared/prerequisites/p22_experiments.md`: baselines, seeds, uncertainty.

Local remediation: lesson C12 states seed and budget before each curve.

U02 Feedback, tools, and constitutional learning

- `../shared/prerequisites/p17_rl.md`: rewards, policies, policy gradients.

Local remediation: lesson C07 defines a reward signal as a scalar attached to a step or an episode before RLEF is discussed.

- `../shared/prerequisites/p20_tools.md`: tool-call/result loop, state,

budgets. Local remediation: lesson C01 restates the ReAct loop as thought, action, observation triples with a typed trace.

- `../shared/prerequisites/p21_security.md`: trust boundaries, least

privilege, prompt injection. Local remediation: lesson C12 treats the independent validator as a trust boundary and lists what it must not see.

U03 Planning, search, and train-time RL

- `../shared/prerequisites/p09_optimization.md`: objectives, constraints,

gradients. Local remediation: lesson C11 frames the compute budget as a hard constraint in the planning objective.

- `../shared/prerequisites/p17_rl.md`: value functions, credit assignment,

on-policy versus off-policy. Local remediation: lesson C05 re-derives multi-step credit from a three-step toy before STaR and GRPO appear.

- `../shared/prerequisites/p20_tools.md`: tool environments and state.

Local remediation: lesson C04 defines a plan node as a state plus a pending subplan before parallel execution is discussed.

U04 Open-ended evolution and deep research

- `../shared/prerequisites/p20_tools.md`: code execution sandboxes.

Local remediation: lesson C12 lists the sandbox rules before any self-modifying loop is designed.

- `../shared/prerequisites/p22_experiments.md`: holdouts, novelty claims,

negative results. Local remediation: lessons C08, C09, C11 teach evidence provenance and holdout separation as first-class mechanisms, not asides.

Diagnostic

`diagnostics/diagnostic.md` probes all four bridge groups with 12 questions. `diagnostics/diagnostic_key.md` carries the key. The mastery ledger records that no learner answers are invented, the diagnostic is a self-check tool.

Not-yet-understood dependency rule

Every lesson file ends with a numbered dependency list for concepts the learner may still lack. The list points at the bridge file and the local remediation section, never at another course alone.

Notation and shapes

Every symbol below is defined before its first use in lessons.

Sampling and verification (U01)

- N : number of independent samples drawn for one prompt. Unit: count.

Shape: scalar integer, $N \geq 1$.

- p : single-sample success probability. Unit: probability in $[0, 1]$.
- $\text{pass}@N$: probability that at least one of N samples is correct.

$\text{pass}@N = 1 - (1 - p)^N$ under independence.

- G : generator policy. Maps prompt x to a distribution over completions y .
- V : verifier. Maps (x, y) to a scalar score s in $[0, 1]$ or $\mathbb{R}$.
- s_i : verifier score of sample i . Shape: scalar.
- B : compute budget. Unit: tokens, FLOPs, or dollars. Always name the unit.
- c_s : cost per sample. Total cost of best-of- N is about $N * c_s$.

Feedback and tools (U02)

- Trace T : ordered list of steps. $T = [t_1, \dots, t_K]$.
- Step t_k : triple $(\text{thought}_k, \text{action}_k, \text{observation}_k)$.
- action_k : tool call with name and typed arguments.
- observation_k : tool return, typed as value or error.
- r_k : reward signal at step k . Scalar. Outcome reward sets $r_k = 0$ for $k < K$ and r_K in $\{0, 1\}$ or $\mathbb{R}$. Process reward allows nonzero r_k for $k < K$.
- Critique $C(y)$: text judgment of output y . No weight change follows.
- Policy update: weight change from data, for example one gradient step.

Planning, search, train-time RL (U03)

- Node n : state plus pending subplan in a search tree.
- b : branching factor. d : search depth. Leaf count of full expansion: b^d .
- π_θ : policy with parameters θ . $\pi_\theta(a | s)$ is the action probability.
- Trajectory τ : $(s_0, a_0, r_0, \dots, s_T)$. Return $R(\tau)$ = sum of rewards.
- Advantage $A(s, a)$: $Q(s, a) - V(s)$. In GRPO, advantage of response i in a group of G responses is $(r_i - \text{mean}(r)) / \text{std}(r)$.
- $\text{KL}(\pi || \pi_{\text{ref}})$: divergence penalty that keeps the policy near a reference policy. Unit: nats.
- On-policy: update data comes from π_θ itself.

Evolution and deep research (U04)

- Population P_t : set of candidate agents at generation t . $|P_t| = M$.
- Fitness $f(c)$: scalar score of candidate c on the validation split.
- Mutation $m(c)$: stochastic edit of candidate c .
- Selection: keep top- k by fitness, or sample proportional to fitness.
- Holdout H : locked test set, never used for selection or tuning.
- Provenance record: (claim, source, retrieval path, date) for every evidence-backed statement a research agent makes.

General

- Seed: integer that fixes all RNG draws in a run. Always reported.
- CI: confidence interval, with the method named (for example Wilson).
- All toy numbers in lessons are computed by hand-traceable arithmetic or by the executed lab scripts. Nothing is a benchmark claim.

Glossary

- Advantage: how much better an action is than the average action in a state.
- AlphaCode: system for competition-level code generation with large-scale sampling, filtering, and clustering. Source attribution pending.
- AlphaEvolve: evolutionary coding agent for algorithm design.

Source attribution pending.

- Architecture search (inference): search over inference-time techniques and their settings for a task. Source attribution pending.
- Best-of-N: draw N samples, return the one with the highest verifier score.
- Calibration: match between stated confidence and observed accuracy.
- Constitutional AI: training with AI-generated feedback guided by a written constitution of principles. Source attribution pending.
- Cost-success curve: task success as a function of inference compute budget.
- Critique: judgment of output that does not change model weights.
- DAPO: open-source LLM reinforcement learning system at scale.

Source attribution pending.

- Decomposition: split a task into subtasks solved separately.
- Execution feedback: signal from running code or a tool, such as a test result or an error trace.
- Generator/verifier gap: difference between the ability to produce a correct answer and the ability to recognize one.
- GRPO: group-relative policy optimization, advantages computed against a group baseline instead of a learned value model. Source attribution pending.

- Holdout separation: a locked test set kept apart from all selection and tuning decisions.
- Independent validation: a separate check with its own data and rules, not tuned on the system it judges.
- LATS: language agent tree search, unifying reasoning, acting, and planning. Source attribution pending.
- On-policy: training data drawn from the current policy.
- Outcome reward: scalar feedback on the final result only.
- Pass@N: probability that at least one of N independent samples is correct.
- Process reward: scalar feedback on intermediate steps.
- ReAct: loop of thought, action, and observation for tool use.

Source attribution pending.

- Reward hacking: the policy exploits a flaw in the reward instead of the intended task. Also called shortcut exploitation.
- RLEF: reinforcement learning from execution feedback.

Source attribution pending.

- Selection bias: systematic error from choosing samples by a biased score.
- Self-consistency: sample several reasoning paths, take the majority answer.
- STaR: bootstrap reasoning by training on self-generated rationales that lead to correct answers. Source attribution pending.
- Test-time compute: inference budget spent per query, such as samples, search steps, or verifier calls.
- Tree search: explore action sequences as a tree of states.

- Verifier: model or rule that scores candidate outputs.
- Weak verifier: a verifier weaker than the generator that still helps selection. Source attribution pending.
- Ablation delta: full-system score minus score with one component removed. Causal, not correlational.
- AlphaGeometry: neuro-symbolic geometry prover: the net proposes constructions, the symbolic engine closes by deduction. Source attribution pending.
- AlphaProof: formal mathematics proving with learning-guided search. Source attribution pending.
- Amdahl's law: $\text{speedup} = 1 / ((1-f) + f/s)$. The unoptimized share caps everything.
- Archival store: MemGPT's cold tier: evicted facts paged out of main context, returned via search.
- Autonomy envelope: authority must not exceed verified capability, with margin.
- Blind-spot question: a question where the generator and a contaminated verifier share the failure mode.
- CacheBlend: reuse non-prefix chunk KV caches. Recompute only high-deviation tokens. Source attribution pending.
- Cartridge: a small trained KV cache per corpus, built by self-study. Source attribution pending.
- CodeMonkeys: SWE-agent scaling: serial iterations raise coverage, parallel width amortizes context. Source attribution pending.
- Contamination gap: same-model judge agreement minus independent-judge agreement.
- DeepScholar-Bench: live benchmark for research synthesis: synthesis, retrieval quality, verifiability. Source attribution pending.
- Deviation check: CacheBlend's scan for high-KV-deviation tokens before selective recompute.
- Duration curve: task success rate versus task duration in hours.
- fast_p: share of kernels both correct and at least p times faster than reference.

- Flaky test: a test that passes or fails nondeterministically.
- Gate: a project milestone with exit criteria and a kill rule.
- GDPVal: benchmark of economically valuable tasks with blinded expert comparison. Source attribution pending.
- Hidden test leakage: the agent sees or infers the hidden tests it is graded on.
- Judge contamination: the judge shares the generator's training and blind spots.
- Kernel (proof): the trusted checker that accepts or rejects a formal proof step.
- KernelBench: benchmark: rewrite a PyTorch reference as a fast GPU kernel. Source attribution pending.
- KV cache: stored keys and values of past tokens. The working memory of inference.
- LMCache: guest-topic KV cache sharing system (schedule line only). Source attribution pending.
- Mean-min flip: the mean picks one agent, the minimum picks the other.
- MemGPT: agent that pages its own memory between main context and archival storage. Source attribution pending.
- Negative result: a preregistered hypothesis rejected with power. Reported, not filed away.
- Optimal stopping: continue while the next step's expected value beats its cost.
- Preregistration: the falsification is written before the run, so the measurement decides.
- Profiling: measuring where time goes before optimizing.
- Reality gap: simulator success minus real-world success.
- Sandboxing: running untrusted agent code in an isolated environment.
- Selection trajectory: the final pick among candidate trajectories in a SWE-agent pipeline.
- Self-study: training a Cartridge on synthetic conversations distilled from a corpus.
- Serial depth: refinement iterations per trajectory in test-time code search.
- Sim-to-real: transferring a policy trained in simulation to the real world.
- Stopping rule: stop when marginal gain falls below marginal cost.

- Tactic: one proof step in a formal proof assistant.
- Task value: hours saved times the hourly rate of that time.
- Trace faithfulness: a trace is faithful iff flipping its cited fact changes the behavior.
- Value capture: the share of total task value the agent's successes represent.
- Verifier-weighted voting: the hybrid: weight each sample's vote by its verifier score.
- Wilson interval: a confidence interval for a binomial rate, used on all toy success rates.

Cheatsheet

One page per unit is the goal. This is the dense reference. Formulas, decision rules, toy numbers, failure modes. Detail in lessons/. The fast pass in `crash-course.md`.

U01 test-time compute

- $\text{pass@N} = 1 - (1-p)^N$. $p=0.25$, $N=5$: 0.76.
- Best-of-N: sample N, keep verifier's top. Needs verifier pairwise accuracy > 0.60 . Below 0.55 it is untestable.
- Self-consistency: sample N, majority vote. Needs per-sample accuracy above chance.
- Matched budget: $N_{sc} \text{ gen} = N_{bon} (\text{gen} + \text{score})$.
- Selection bias: check score-vs-length correlation.
- Cost-success curve is concave: read it before allocating.

U02 feedback and tools

- ReAct: thought $\rightarrow$ action $\rightarrow$ observation $\rightarrow$ answer.
- Execution feedback = ground truth. Self-critique = opinion.
- Tool error: read it, fix, retry once. Then replan.
- Shortcut: the agent edits the test. Catch with independent validation (separate judge, locked tests).
- Critique changes the try. A gradient changes the weights.

U03 planning and search

- Tree search: state, action, reward. Random hit rate:
 $\text{correct-leaves} / \text{branching}^{\text{depth}}$.
- Perfect verifier: free node labels. Learn only the policy.
- Reward validity: correlate reward with truth on held-out traces before trusting search.
- Budget binds: fix budget, then tune. Never the reverse.

U04 open evolution

- Loop: vary, select, inherit. Selection pressure = fraction that reproduces.
- Deceptive proxy: selection grows what it measures.
- Holdout rule: champion scored on unseen data with the true metric. Gap = deception meter.
- Kill rule: a gate without teeth is theater.

U05 SWE and kernel agents

- Coverage: 0.832 ($S=1$) to 0.997 ($S=5$) at $K=8$. Depth buys coverage. Width amortizes context (25.0% $\rightarrow$ 7.7%).
- Selection: generate, vote with generated tests, final trajectory.
- fast_p : correct AND $\geq p \times$ faster. Toy: correctness 0.60, fast_1 0.40.
- Amdahl: $\text{speedup} = 1 / ((1-f) + f/s)$. 2x on 60%: 1.43x.
Ceiling $1/(1-f) = 2.5x$.

U06 memory and caches

- CacheBlend: recompute high-deviation tokens only. Toy:

2048 -> 307 + 64 check = 5.52x.

- MemGPT: 8 hot slots, 12 paged to archival, 5-search recall

index. Evict the coldest. Search returns them.

- Cartridge: trained KV per corpus. 20,480 -> 512 slots

(40x). Training amortizes over queries.

U07 reasoning, formal, autonomy

- Proof search: goal, tactic, kernel. Random $(1/12)^4$.

guided $(1/3)^4$.

- AlphaGeometry: net proposes, engine deduces.
- Report search nodes and proof steps separately.
- Reality gap: sim - real. Randomize to narrow it.
- Envelope: authority $\leq$ verified capability + margin.
- Faithful trace: flip the cited fact. Behavior must change.

U08 long-horizon eval

- Duration curve: success vs hours. Toy: 0.90 -> 0.35.
- Value = hours x rate. Score where the money is.
- GDPVal: blinded pairwise win rate. $SE = \sqrt{p(1-p)/n}$.
- DeepScholar: synthesis / retrieval / verifiability.
- Ship on the min, not the mean.
- Stop: continue while marginal value > marginal cost.

- Contamination gap = same-model agreement - independent agreement.
- Budget fuels: agent + judge + audit. The audit binds.
- Ablation delta = full - removed. Causal, not correlational.

Cross-unit rules

- Generate candidates, verify cheaply, select honestly.
- The verifier/judge is part of the apparatus: calibrate it, measure its bias, keep it at a distance.
- Report the tail (min), the cost (budget), and the negative (null with power).
- A gate without a kill rule is theater. A metric without a falsification is a wish.

Crash course

The whole course in one pass: 8 units, 96 concepts, the mechanisms that matter and the numbers that anchor them. Detail lives in `lessons/u01-u08`. Practice in `labs/`, `interview/`, `capstones/`.

The arc

U01 spends compute at test time. U02 grounds the agent with tools and feedback. U03 plans and searches. U04 evolves agents. U05 builds software-engineering and kernel agents. U06 gives them memory. U07 reasons formally and bounds autonomy. U08 evaluates long-horizon work and turns the course into research projects.

U01 , Test-time compute and verification

More tries help when tries differ. $\text{pass@N} = 1 - (1-p)^N$: with $p=0.25$ and $N=5$, the set wins 0.76 of the time while one ticket wins 0.25. Best-of-N samples N and keeps the verifier's top pick. Self-consistency samples N and takes the majority. The generator/verifier gap: verification is cheaper than generation, which is why selection works. Weak verifiers help only above 0.60 pairwise accuracy (below 0.55 the method is untestable). Selection bias is the failure: a verifier that rewards length selects for length. The cost-success curve is concave: each new sample adds less. Read it to allocate a fixed budget.

U02 , Feedback and tools

ReAct: thought, action, observation, repeat. Execution feedback is ground truth from the world. Self-critique is the model's opinion and shares its blind spots. Code and tests are the tightest loop: the test suite is a semi-trusted judge. Critique is not a learning update: a comment changes the next try, a gradient changes the weights. Constitutional feedback is critique with principles. Reward signals shape behavior. Constraints bound it. Tool errors are data: read the traceback, do not repeat the call. Shortcut exploitation is the agent doing the wrong thing correctly (editing the test). Independent validation is the separate judge that execution feedback cannot be.

U03 , Planning and search

Tree search: state (reasoning so far), action (next step), reward (right answer). A perfect verifier makes search tractable: every node is labeled free, so only the policy is learned. Adaptive branching spends compute where it matters. Decomposition splits the problem. Parallel planning runs branches together. Multi-step credit is the hard part: which step caused the win. Synthetic traces, STaR, and reasoning RL all manufacture training signal from search. DAPO/GRPO are the reading pointers. Stay on-policy: train on what you deploy. The budget binds everything. Reward validity is the last word: a high reward that does not mean a right answer is a hack, not a reward.

U04 , Open-ended evolution

Vary a population of agent architectures, select the fittest, inherit with mutation. Novelty search escapes deceptive landscapes by rewarding difference. Selection is Goodhart in a population: it selects on the proxy, so the proxy is what grows. Holdout separation is the law: the champion is scored on data the search never saw, with the true metric. Evidence provenance tracks where every number came from. Resource budgets bound the search. Novelty claims need the holdout gap. Safe iteration means the kill rule is real: a project that cannot pass a gate dies at the gate.

U05 , Software-engineering and kernel agents

CodeMonkeys: serial iterations S raise coverage (0.832 at $S=1$ to 0.997 at $S=5$ with $K=8$), parallel width K amortizes the shared context (context share 25.0 percent at $K=8$, 7.7 percent at $K=32$). Selection has three stages: generate, vote with model-generated tests, and a final selection trajectory. KernelBench: the model rewrites a PyTorch reference as a GPU kernel. `fast_p` counts kernels that are both correct AND at least p times faster (correctness 0.60 but `fast_1` 0.40 on the toy: speed without correctness scores zero). Amdahl: 2x on 60 percent gives 1.43x, never more than 2.5x. Profile first: the biggest share sets the ceiling.

U06 , Memory, caches, and long-context

The KV cache is the working memory of inference. CacheBlend reuses non-prefix chunk caches and recomputes only high- deviation tokens: 2,048 prefill units become 307 recomputed plus 64 of deviation check, a 5.52x saving. MemGPT pages its own memory: hot facts stay in main context (8 slots), cold facts are evicted to archival storage (12 in the toy) and return via search. 20 turns, 27 memory actions, nothing lost. A Cartridge trains a small KV cache per corpus by self-study: 20,480 tokens distill into 512 trained slots, 40x fewer per query. Training amortizes over queries.

U07 , Reasoning, formal systems, and autonomy

Formal proof search: goal (what remains), tactic (a step), kernel (the checker). The kernel is the free verifier: every node is labeled right/wrong for free. AlphaGeometry's split: the neural net proposes constructions (ideas are cheap), the symbolic engine closes by deduction (deduction decides). Report search and proof separately: 13 nodes searched, 4-step proof. Sim-to-real: domain randomization narrows the reality gap (0.25 to 0.12 on the toy). The autonomy envelope: authority must not exceed verified capability, with margin. 3 of 4 configs admitted. The low-capability high-authority cell rejected. Traces: a trace is faithful iff flipping the cited fact changes the behavior. 30 percent of plausible traces are stories.

U08 , Long-horizon evaluation and research projects

Duration is difficulty in disguise: 0.90 at 15 minutes, 0.35 at 8 hours on the toy. Short-task scores do not extrapolate. Value = hours times hourly rate: weight tasks by value, not count. GDPVal: experts from 44 occupations write real deliverables. Blinded pairwise comparison gives the win rate. 220 gold tasks anchor it. DeepScholar-Bench grades research synthesis in three dimensions: synthesis, retrieval quality, verifiability. The split diagnoses the system (fluent 0.80 but thin 0.35). Ship on the minimum, not the mean: A means 0.767 but mins 0.55, B means 0.72 and mins 0.70. Stop when the next hour's value falls below its cost. The contamination gap: same-model judge 0.85 minus independent judge 0.60 = 0.25 of similarity, not quality. Budget three fuels: agent, judge, audit. The audit binds. Ablate: the delta is causal. Gates with kill rules. Negative results with power. Five open questions.

How the units connect

U01's verifier is U07's kernel pushed to the limit (infallible) and U08's judge pushed to the real world (fallible). U02's execution feedback is U03's reward signal grounded. U04's holdout separation is U08's eval hygiene at population scale. U05's selection stage is U01's best-of-N with tests as the verifier. U06's paging is U03's search over memory. The course is one idea restated eight ways: generate candidates, verify cheaply, select honestly, and report what the verifier cannot see.